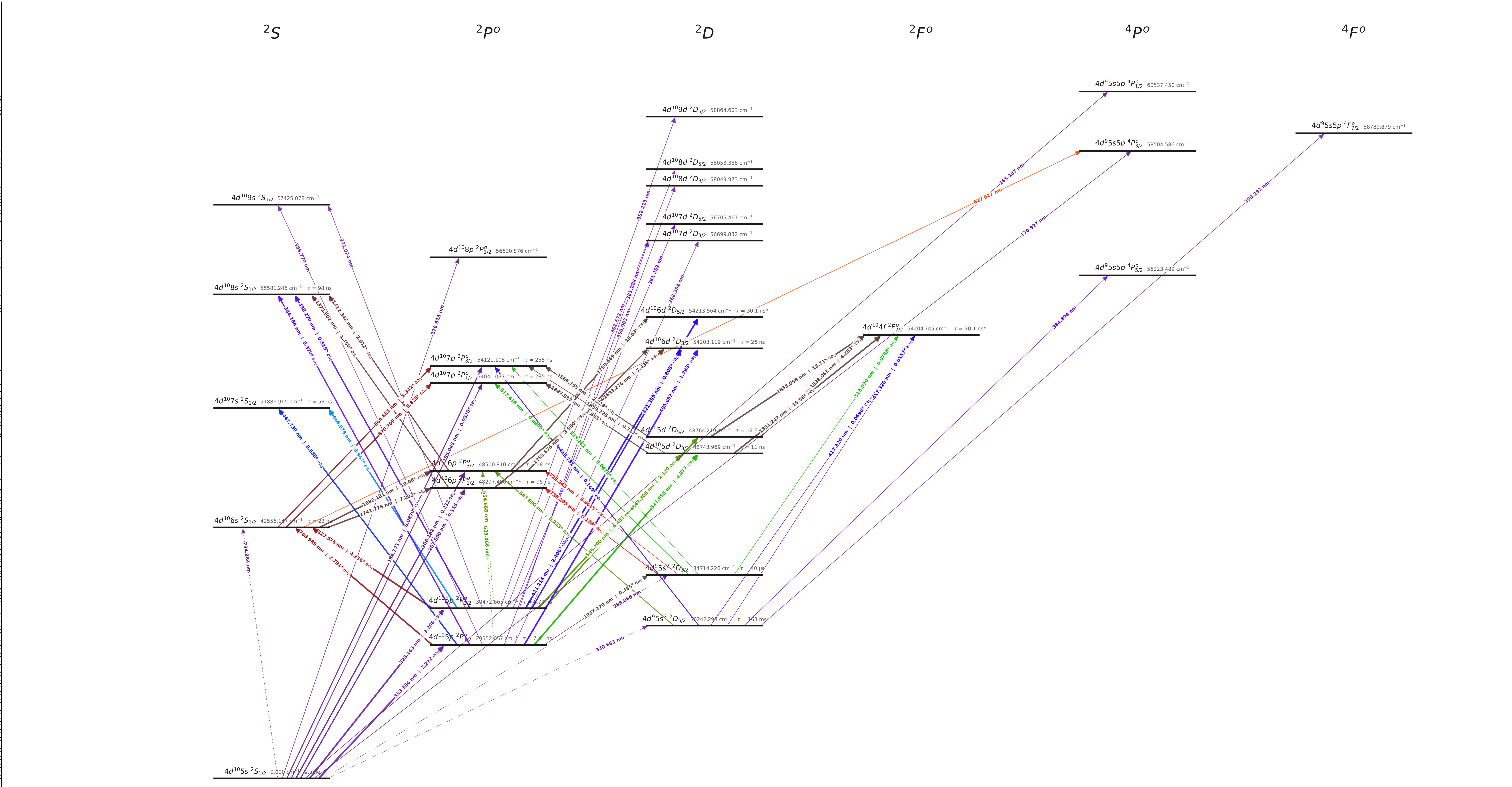


Silver

Ag (Ag I). Strongest classified transitions below 2 μm (NIST): 29 levels, 62 lines.

Ag⁺ ionisation limit 61106.45 cm⁻¹ = 7.57623 eV (163.649 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[<i>J er J</i>] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{3/2}	288.0663	287.9818	1040.7063	–	1.130e-05	0.003979	0	two-photon line at 576 nm
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{3/2}	328.1625	328.0680	913.548766(20)	3.206	1.473e+08	23.44	100	D2 resonance line
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{5/2}	330.6628	330.5676	906.641024	–	5.220e+00	9.764e-07	85.1	ultra-narrow E2 clock transition
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{1/2}	338.3859	338.2888	885.9484	2.272	1.350e+08	21.48	100	D1 resonance line
<i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{5/2} – <i>4d</i> ¹⁰ <i>6p</i> ² <i>P</i> _{3/2}	547.6898	547.5376	547.3764	0.233*	1.668e+05	2.04	1.3	547.7 nm line used for spectroscopy of the metastable 2D5...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

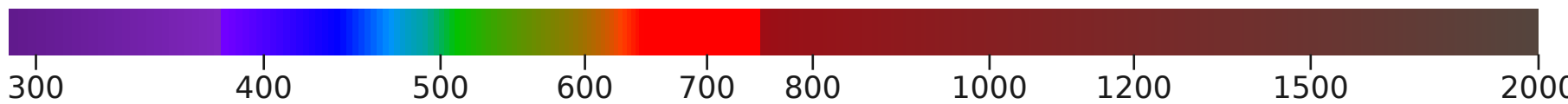
Arrow label: vacuum wavelength | reduced dipole matrix element [*J||er||J*] in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



¹⁰⁷Ag hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2}	-1712.512111(18)	–	F=0: +1284.38 F=1: -428.13
<i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{1/2}	-175.4(17)	–	F=0: +131.55 F=1: -43.85
<i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{5/2}	-126.28180(10)	–	F=2: +220.99 F=3: -157.85
<i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{3/2}	-31.7(5)	–	F=1: +39.62 F=2: -23.77
<i>4d</i> ¹⁰ <i>6p</i> ² <i>P</i> _{1/2}	-38.7(10)	–	F=0: +29.02 F=1: -9.68
<i>4d</i> ¹⁰ <i>6p</i> ² <i>P</i> _{3/2}	-9.05(25)	–	F=1: +11.31 F=2: -6.79
<i>4d</i> ¹⁰ <i>7p</i> ² <i>P</i> _{3/2}	-4.50(20)	–	F=1: +5.62 F=2: -3.38

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>6p</i> ² <i>P</i> _{3/2}	206.182	107-109	414.0(6)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>6s</i> ² <i>S</i> _{1/2}	234.984	107-109	368(7)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{3/2}	328.163	107-109	473.2(7)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{5/2}	330.663	109-107	564.1(4)
<i>4d</i> ¹⁰ <i>5s</i> ² <i>S</i> _{1/2} – <i>4d</i> ¹⁰ <i>5p</i> ² <i>P</i> _{1/2}	338.386	107-109	473(4)
<i>4d</i> ⁹ <i>5s</i> ² ² <i>D</i> _{5/2} – <i>4d</i> ¹⁰ <i>6p</i> ² <i>P</i> _{3/2}	547.690	109-107	-978.1(5)

Shift = ν(first isotope) – ν(second isotope).